MERN STACK

DAY – 37

6 August 2026

PROJECT: PARKEASY - PARKING MANAGEMENT SYSTEM

BOOKING MANAGEMENT

• The Booking Management module is developed to allow administrators to manage all parking booking requests submitted by users.
• It provides a centralized interface for viewing booking details, approving requests, rejecting requests, and monitoring completed parking records.
• The module is implemented using React.js for the frontend and Express.js with MongoDB for backend processing.
• The useState() hook is used to store all booking records retrieved from the server.
• The useEffect() hook automatically loads booking data when the Booking Management page is opened.
• The Sidebar component is integrated to provide navigation within the administrator panel.
• The booking information is fetched dynamically from the backend using REST APIs.

Booking Management APIs

GET /all-bookings

• Retrieves booking records of all users from the MongoDB database.
• Combines booking information with user details such as name, vehicle number, slot number, booking date, booking time, and current booking status.
• Returns a complete list of booking requests for display in the administrator dashboard.

PUT /booking-status/:userId/:index

• Updates the status of a selected booking request.
• Allows administrators to approve or reject pending booking requests.

• If a booking is rejected or cancelled, the related parking slot is made available again.
• Returns a confirmation message after successfully updating the booking status.

Features

• Displays all booking requests submitted by users.
• Organizes bookings into separate sections such as Pending Requests, Approved Bookings, Completed Parking, and Cancelled/Rejected Bookings.
• Displays user name, parking slot, booking date, booking time, vehicle number, and booking status.
• Allows administrators to approve or reject pending booking requests.
• Automatically refreshes booking records after every update.
• Displays appropriate booking status for each request.

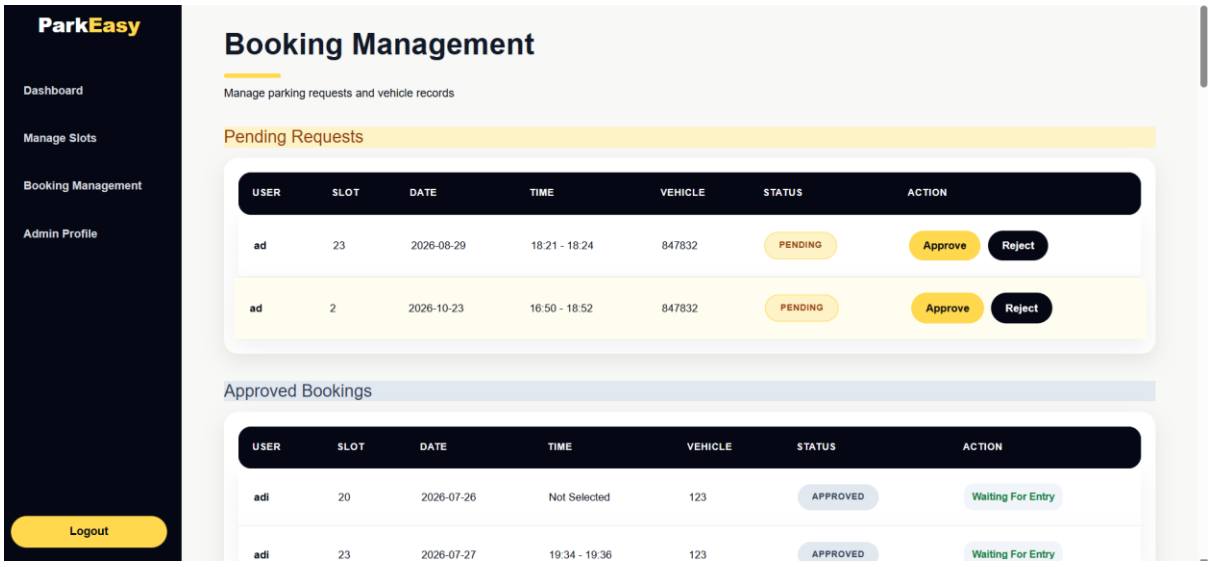
Working

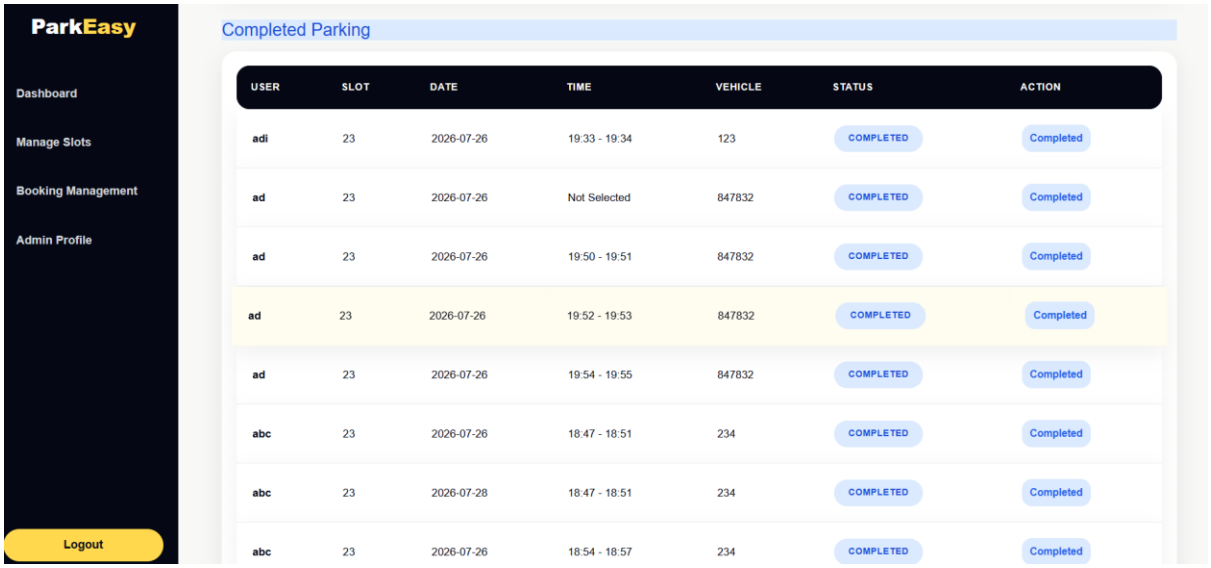
1. The BookingManagement() component is created to manage all parking booking requests.
2. The component retrieves booking information using the GET /all-bookings API when the page loads.
3. The backend collects booking records from every registered user and prepares a combined booking list.
4. The received booking data is stored using the useState() hook and displayed in different booking sections.
5. Pending booking requests display Approve and Reject buttons for administrator actions.
6. When an administrator performs an action, the PUT /booking-status/:userId/:index API is called to update the booking status.
7. The backend updates the booking record and makes the parking slot available again if the booking is rejected or cancelled.

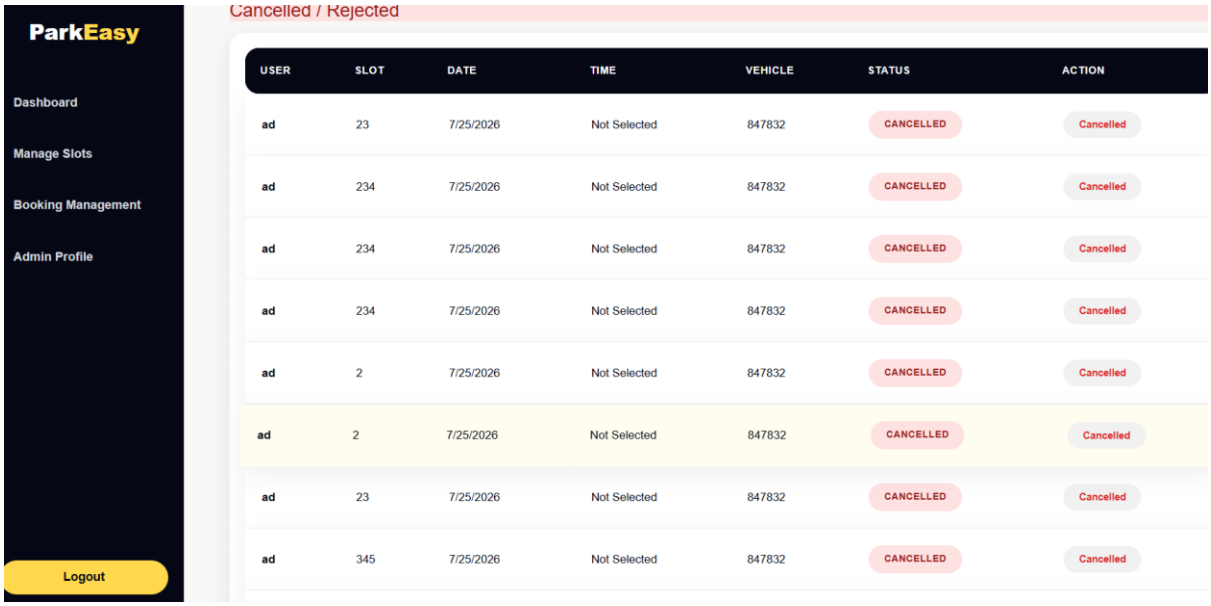
8. After every successful update, the booking list is refreshed automatically to display the latest booking information.
9. Completed and approved bookings are displayed with their respective status labels for easy monitoring.
10. The Booking Management module enables administrators to efficiently control booking requests, monitor parking activities, and maintain smooth parking operations within the ParkEasy system.

```
app.get("/all-bookings", async(req,res)=>{
try{
const users=await User.find();
let bookings=[];
users.forEach(user=>{
user.bookings.forEach((booking,index)=>{
bookings.push({
user:user.name,
userId:user._id,
bookingIndex:index,
slot:booking.slot,
date:booking.date,
time:
booking.startTime && booking.endTime
?
`${booking.startTime} - ${booking.endTime}`
:
booking.time || "Not Selected",
vehicle:
booking.vehicle ||
user.vehicle?.number ||
"Not Added",
status:
booking.status || "Pending"
});
});
});
res.json(bookings);
}
catch(err){
res.status(500).json({
message:err.message
});
}
});
```

```
app.put("/booking-status/:userId/:index",async(req,res)=>{
try{
const {status}=req.body;
const user=await User.findById(req.params.userId);
if(!user){
return res.json({
success:false,
message:"User not found"
});
}
const index=Number(req.params.index);
const booking=user.bookings[index];
if(!booking){
return res.json({
success:false,
message:"Booking not found"
});
}
booking.status=status;
if(status==="Approved"){
booking.status="Approved";
// DO NOT CHANGE SLOT STATUS
}
if(status==="Rejected" || status==="Cancelled"){
await Slot.findOneAndUpdate(
{
slotNumber:booking.slot
},
{
status:"Available",
bookedBy:null
}
);
}
await user.save();
res.json({
success:true,
message:`Booking ${status}`
});
}
catch(err){
res.status(500).json({
success:false,
message:err.message
});
}
});
```




ParkEasy
Dashboard
Manage Slots
Booking Management
Admin Profile
Logout
Completed Parking
USER SLOT DATE TIME VEHICLE STATUS ACTION
adi 23 2026-07-26 19:33 - 19:34 123 COMPLETED Completed
ad 23 2026-07-26 Not Selected 847832 COMPLETED Completed
ad 23 2026-07-26 19:50 - 19:51 847832 COMPLETED Completed
ad 23 2026-07-26 19:52 - 19:53 847832 COMPLETED Completed
ad 23 2026-07-26 19:54 - 19:55 847832 COMPLETED Completed
abc 23 2026-07-26 18:47 - 18:51 234 COMPLETED Completed
abc 23 2026-07-28 18:47 - 18:51 234 COMPLETED Completed
abc 23 2026-07-26 18:54 - 18:57 234 COMPLETED Completed


ParkEasy
Dashboard
Manage Slots
Booking Management
Admin Profile
Logout
Cancelled / Rejected
USER SLOT DATE TIME VEHICLE STATUS ACTION
ad 23 7/25/2026 Not Selected 847832 CANCELLED Cancelled
ad 234 7/25/2026 Not Selected 847832 CANCELLED Cancelled
ad 234 7/25/2026 Not Selected 847832 CANCELLED Cancelled
ad 234 7/25/2026 Not Selected 847832 CANCELLED Cancelled
ad 2 7/25/2026 Not Selected 847832 CANCELLED Cancelled
ad 2 7/25/2026 Not Selected 847832 CANCELLED Cancelled
ad 23 7/25/2026 Not Selected 847832 CANCELLED Cancelled
ad 345 7/25/2026 Not Selected 847832 CANCELLED Cancelled